# Ideas and structure for the contributed talk.

Biological tissues are highly anisotropic, meaning that their diffusion rates are not the same in every direction. For routine DW imaging we often ignore this complexity and reduce diffusion to a single average value, the apparent diffusion coefficient (ADC), but this is overly simplistic. A superior method to model diffusion in complex materials is to use the diffusion tensor, a 3 x 3 array of numbers corresponding to diffusion rates in each combination of directions. The three diagonal elements (Dxx, Dyy, Dzz) represent diffusion coefficients measured along each of the principal (x-, y- and z-) laboratory axes. The six off-diagonal terms (Dxy, Dyz, etc) reflect reflect the correlation of random motions between each pair of principal directions.

Coriolis terms

Rescaling.

The Gronwall part.

Are these weak solutions and not L-H WS.

trace and boundary. h -1 over 2

coercivity of the diffusion matrix and the eigenvalue.

why the $\epsilon^{-3}$ in the approximated function.

any comment on the lateral conditions?

IMPORTANT: u3 BCs for the limit model case. $un_3 = 0$

weak formulation: the form itself comes from calculations, but the regularities $L^\infty L^2$ and $L^2 H^1$ come simply from definition, not from necessity of the terms, right? For example, $L^\infty$ regularity is not needed explicitly, is it? The regularities for the test function on the other hand come from what is needed for the expressions to make sense.

why do we need $\tilde{u}_H$ to be in $H^1$ in time? regularities in the weak formulation.

stating the main theorem why can we say that $C_0 = 0$ on the boundary, when it is only an $L^2$ function on the domain?

stating the theorem — $u_0$ is there the initial value of $u_\epsilon, u$, or both? —- It is the initial value belonging to the anisotropic system. It is needed at the energy inequality.

the definition of weak solutions: the u vector is in V. then we conclude at some point, $u_3$ is in $L^2 L^2$... but it was by definition in V, which is essentially $H^1$. clear this. Proposition1.

Definition3 - $U = (\mathbf{v}, C)$ instead of $\mathbf{v}$ I think it is $\mathbf{u}_H$.

OK: Existence theorem - p25 in the Temam article - it states that the solution is in $V$, which is a full $H^1$ space? Yes but in the $U = (u_H, S, T)$ there is no $u_3$ term.

What is the main difference in the existence theorem, comparing their result and the one for $C$: a) we allow $\delta$ to be the source term, not just $L^2$ functions, b) the definition of the $b$ term in our case is different, we have the derivative on the test function, not on the second term. Is there a reason why in their case they keep the derivative on the original function in some cases and not put it onto the test function? Does our case benefit / get some more generality with this?